



Department of Electrical and Electronic Engineering

Name of the Experiment: Verification of Kirchoff's law

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Date of Experiment: 26-11-25

Date of Submission: 03-12-25

Signature
3/12/25

Theory:

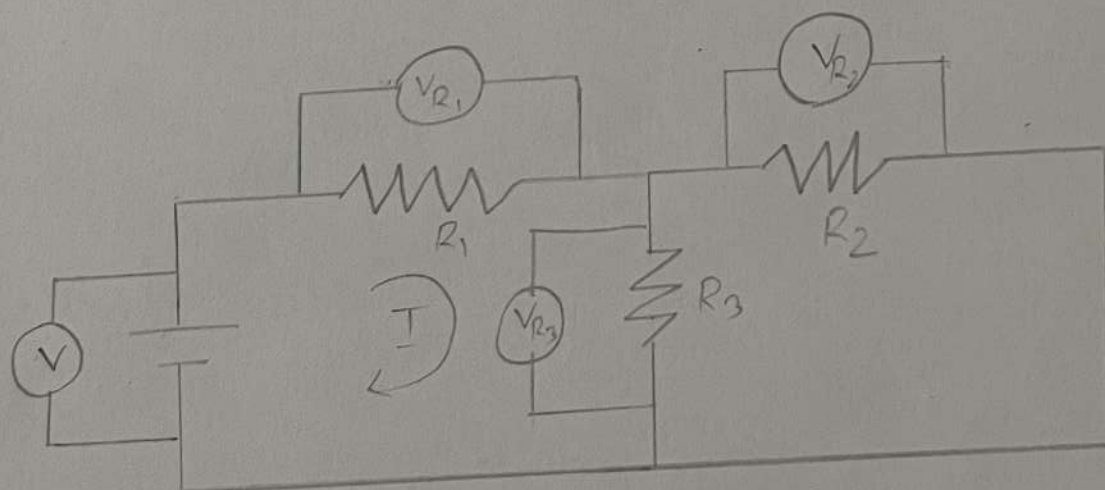
KCL: The algebraic sum of the currents meeting at a node is equal to zero

KVL: In any closed loop/mesh, the algebraic sum of all the voltages is zero.

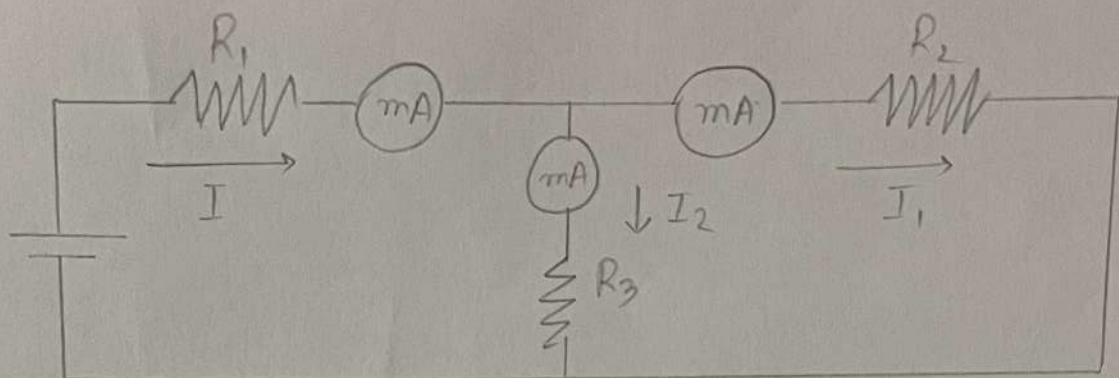
Equipments:

1. Variable DC power supply - 1 piece.
2. Voltmeter and ammeter
3. Resistances
4. Bread Board
5. Connecting wires.

Circuit Diagram:



Verification of KVL



Verification of KCL

Calculation:

$$R_{eqn} = R_1 + (R_2 \parallel R_3) = 1000 + \left(\frac{1}{100} + \frac{1}{100} \right)^{-1} = 1050 \Omega$$

$$V_{R_1} = V \times \frac{R_1}{R_{eqn}}$$

$$V_{R_2} = V_{R_3} = V \times \frac{R_2 \parallel R_3}{R_{eqn}}$$

$$I_1 = I \times \frac{R_3}{R_2 + R_3}$$

$$I_2 = I \times \frac{R_2}{R_2 + R_3}$$

KVL:

For 10V,

$$V_{R_1} = 10 \times \frac{1000}{1050} = 9.52V$$

$$V_{R_2} = 10 \times \frac{50}{1050} = 0.476V$$

$$\therefore V_{R_1} + V_{R_2} = 10V$$

For 15V,

$$V_{R_1} = 15 \times \frac{1000}{1050} = 14.28V$$

$$V_{R_2} = 15 \times \frac{50}{1050} = 0.72V$$

$$\therefore V_{R_1} + V_{R_2} = 15V$$

For 20V,

$$V_{R_1} = 20 \times \frac{1000}{1050} = 19.05 \text{ V}$$

$$V_{R_2} = 20 \times \frac{50}{1050} = 0.95 \text{ V}$$

$$\therefore V_{R_1} + V_{R_2} = 20 \text{ V}$$

KEL:

$$\text{For } 5\text{V}, I = \frac{5}{1050} = 0.0047619 \text{ A}$$

$$\therefore I_1 = 0.0047619 \times \frac{100}{200} = 2.381 \text{ mA}$$

$$\therefore I_2 = 0.0047619 \times \frac{100}{200} = 2.381 \text{ mA}$$

$$\therefore I_1 + I_2 = 4.7619 \text{ mA}$$

For 10V,

$$I = \frac{10}{1050} = 9.5238 \text{ mA}$$

$$\therefore I_1 = 9.5238 \times \frac{100}{200} = 4.7619 \text{ mA}$$

$$\therefore I_2 = 9.5238 \times \frac{100}{200} = 4.7619 \text{ mA}$$

$$\therefore I_1 + I_2 = 9.5238 \text{ mA}$$

For 15V,

$$I = \frac{15}{1050} = 14.2857 \text{ mA}$$

$$\therefore I_1 = 14.2857 \times \frac{100}{200} = 7.143 \text{ mA}$$

$$\therefore I_2 = 14.2857 \times \frac{100}{200} = 7.143 \text{ mA}$$

$$\therefore I_1 + I_2 = 14.2857 \text{ mA}$$

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2024-3-60 -082

Experimental Data: (KVL)

R ₁	R ₂	R ₃
1000 Ω	100 Ω	100 Ω

SI No.	Supply Voltage V (V)	V _{R1} (V)	V _{R2} (V)	V _{R3} (V)	Algebraic sum of voltages (V) V _{R1} + V _{R3} [Measured value]	Algebraic sum of voltages (V) V _{R1} + V _{R3} [Theoretical value]
1	10	9.48V	0.46V	0.46V	9.94 V	10 V
2	15	14.19V	0.70V	0.70V	14.89 V	15 V
3	20	18.93V	0.93V	0.93V	19.86 V	20 V

Experimental Data: (KCL)

R ₁	R ₂	R ₃
1000 Ω	100 Ω	100 Ω

SI No.	Voltage (V)	Current (I)	Current (I ₁)	Current (I ₂)	I ₁ + I ₂ (Measured value)	I ₁ + I ₂ (Theoretical value)
1	5	0.45 mA	0.25 mA	0.25 mA	0.7 mA	4.76 mA
2	10	0.98 mA	0.44 mA	0.44 mA	1.42 mA	9.52 mA
3	15	1.4 mA	0.70 mA	0.70 mA	2.1 mA	14.28 mA

26/11/24

Description:

In this experiment, we used the KCL and KVL laws to check how voltage and current divide in a simple circuit.

We calculated V_{R_1} and V_{R_2} using the voltage divider rule, and then added them to compare with the supply voltage.

We also used the current divider rule to find I_1 and I_2 ,

and added them to compare with the total current.

The calculated values matched the KCL and KVL rules, which proves that the laws work correctly in the circuit.